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Homework 1

1. What tool was used:

Gemini

2. What it was used for:

I used Gemini to help brainstorm user stories for different personas using ParkMobile and SpotHero.

3. The prompt(s) used:

Prompt 1:

https://app.parkmobile.io/venues/atlanta-symphony-hall/events/297349

https://spothero.com/search?kind=event&id=1375435&view=dl

Look at these different parking sites and help me brainstorm user stories for Parking facility operators City transportation departments System administrators Finance and billing personnel Mobile application users Third-party payment providers

4. What I kept vs. What I changed
    - **Kept**: I kept some of the backend/admin stories (12,13,18,19,20,22)
    - **Modified**: I adapted the AI's automated partial refunds story into my billing specialist story (16)
    - **Modified:** I came up with license plate checking, however, after looking at the ai response I changed the user from "parking facilities operator" to "enforcement officer" as I found it more fitting
    - **New:** I wrote stories 1-11, 17, 21. Story 21 is inspired by Story 20's token and these references:
      https://techcrunch.com/2024/10/08/google-partners-with-spothero-convenient-parking-reservations-maps-search/
      https://developers.google.com/maps/api-security-best-practices

Homework 2

1. What tool was used:

Gemini

2. What it was used for:

I used Gemini to help generate a three-level WBS for the project. I also used it to see how long tasks should last and to include QA testing for every phase.

3. The prompt(s) used:

    - "Create a starting base for a work break down structure that is no more than three levels for a website and app that helps with parking reservations. Here is an example of what the format should look like:"

    - "Let's add more in this order: 1. General & Project Management (think project initiation) 2. Analysis & Detailed design (scope diagram, DFD, User stories) 3. Frontend 4. Backend 5.Database & Infrastructure 6. Testing 7. Project Closure"

    - "Within this srs template, where can I place my user stories and my work break down structure"

    - "For the work breakdown structure what is a good timeline for a project of this type"

    - "Can you give me the amount of days it will take for each task to take. For example I have the project started on 14th of september and I have it so phase one lasts between september 14 to september 28 making it 11 days on project."

    - "Is there a way to have testing in every phase?"

4. What I kept vs what I changed:
    - **Kept:** I kept the structure, task names, and duration estimates exactly as was generated. I typed it into Microsoft Project to create my Gantt chart.
    - **Modified**: I adjusted some of the task names within Microsoft Project for formatting purposes. I also set up the project start date as September 14, 2026, in MS Project while keeping the AI's standard working day calculations.

Homework 3

1. What tool was used:

Gemini

2. What it was used for:

I used Gemini to help brainstorm and structure a Sprint 1 backlog. I used it to create specific c task cards divided into categories. I also used it to figure out a color-coded tagging that will be implemented later on.

3. The prompt(s) used:
    - "Create a scrum board for the sprint 1 backlog. I am using Trello. Include 5 items for the login, 15 items for the UI for both the operator and user, 10 items for reporting for operator and user."
    - "Should these be the column headings? I do not want this to overpopulate the to-do section"
    - "Should I create a different ones for In Progress and Done columns? also can you tailor the items for a parking reservation app and website please. Here are the user stories to better tailor the items"
    - "I meant have columns like In progress: Login, In progress: User & Operator UI, etc."
    - "I forgot but can you re do the items and add 15 items for backend processes for operator and user"
    - "There are colored labels in Trello that I can use which items can have what labels and what labels should I add"
    - "I am creating a scrum board on Jira for a parking reservation app. The backlog for the parking app should be login (min 5 items), UI (15 items), the backend processes for operator and use (min 15 items), reporting for operator and user (min 10 items). I also need a sprint 1 backlog on this same board"
4. What I kept vs. What I changed
    - **Verbatim / Kept:** I kept the task breakdown for Login (5 items), UI (15 items), Reporting (10 items), and Backend Processes (15 items) generated by the AI after verifying they matched my project's requirements. I also used the recommended Trello column structure (categorized To-Do columns, a unified In Progress section, and Code Review/Testing) and the persona-based color labeling system (e.g., Green for Mobile User, Blue for Operator, Red for Backend) exactly as suggested.
    - **Modified / Changed:** While one of my prompts asked for Jira formatting, I ultimately applied it to Trello. I manually set up the Trello board,

created the cards, and applied the labels myself based on the AI's blueprint.

Homework 4

1. What tool was used:

Gemini

2. What it was used for:

I used Gemini to help brainstorm a 16-entry Risk Register for the Smart Parking Platform, covering technical, schedule, financial, and people risks. I also used it to help guide me on conditional formatting Google Sheets. Finally, I used it to create new user stories based on these newly identified project risks.

3. The prompt(s) used:
    - "Look at these four pdf as context for what you need to do: Using excel I need to create a Risk Register structured as a list with at least 10 entries. It must cover four main categories: technical, schedule, financial, and people. For each risk entry, provide the following specific details: - ID and Description - Probability and Impact - Owner and Response Strategy (specify whether to Avoid, Mitigate, Transfer, or Accept) - Status/Notes" There must also be at least 4 risks for each category."
    - "How can I get it so that any risk labeled as high is red, medium is orange, and low is green?"
    - "Should response strategy be color coded as well?"
    - "Can you add a ranking like as seen on the screenshot? The ranking should rank based on the impact. Furthermore, I would like for the spreadsheet to be formatted so that others are able to see select rows such categories and owners."
    - "2. Add the "Ranking" Column... I wanted it as a numbered ranking instead"
    - "No each row should have their own ranking. Every row with a impact high should have a greater ranking than medium, medium should have a higher ranking than low"
    - "- I would like to separate these two: For status I would like to have these categories: Open / Identified, In Progress / Under Review, Mitigated/Controlled, Occurred / Realized, and Closed / Retired. Please look at the list and just randomly apply them when applicable."
    - "based on these new risks what are new user stories that can occur"
    - "Reformat so it is more similar to this format"
4. What I kept vs. what I changed:

- **Verbatim / Kept:** I kept the 16 specific risks generated for the Risk Register, including their IDs, descriptions, and response strategies, which I inputted into my spreadsheet.
- **Modified / Changed:** I manually applied the suggested Google Sheets conditional formatting and dropdown data validation rules myself. I also kept the newly generated user stories (numbers 23 through 29) exactly, but I changed the formatting so it is more uniformed with the previous user stories.

Homework 5

1. **What tool was used:** Google Gemini
2. **What it was used for:** I generated a RACI matrix, a bottom-up Resource Plan, a Cost Plan with a 10% contingency reserve, a Cashflow Table, and a recommended technology stack for the Smart Parking Platform semester project.
3. **The prompt(s) used:**
    - "Use these files as context and using a bottom-up estimation approach generate the following 1. RACI Matrix Create a RACI matrix (Responsible, Accountable, Consulted, Informed). Include the following roles: developers, project manager, Quality assurance UX, DevOPs, Business analysts, product owner, and Subject matter expert. Map them against the following tasks: Requirements, Design, Implementation, Testing, and Deployment. 2. Resource Plan Create a bottom-up estimation table with the following columns: Role, Name, FTE %, Estimated Hours, Hourly Rate, and Cost. - use the roles listed above. - Assign reasonable hourly rates and estimate realistic hours 3. Cost Plan Create a table with the columns: Item, Category, Qty/Months, Unit Cost, Duration (months), Total, and Notes. Include... total Labor costs from the Resource Plan. - Infrastructure/Cloud hosting costs. - Licenses/Tools costs. Calculate and add a [Insert %, e.g., 10%] Contingency reserve applied to the subtotal before contingency. 4. Cashflow Table Create a cashflow table outlining the spending timeline with the columns: Month, Labor, Infra, Licenses, Contingency, and Total. Distribute the costs from the Cost Plan logically across the project's months. Ensure all calculations are accurate, come up with the appropriate tech for this project as well"
    - "can you make it so that i can copy the cost plan table"
    - "no like create a table and let me copy it"
4. What I kept vs. what I changed:
    - **Verbatim / Kept:** I kept the financial estimates, labor hour calculations, and hourly rates generated by the AI for the Resource Plan and Cost

Plan. I also kept the Cashflow Table's monthly distributions and the role assignments mapping in the RACI matrix.

- **Modified**: I transferred the tables that were generated onto google sheets and color coded the RACI table through conditional formatting. I also formatted the tables for Resource Plan, Cost Plan, and Cashflow Table.

Homework 6

1. **What tool was used:** Google Gemini
2. **What it was used for:** I used ai to identify discrepancies in my project files, format my user stories into formal Software Requirements Specification (SRS) statements (REQ-FUNC, REQ-SEC, REQ-PERF, etc.), and I also used it to get advice on where to place my assignments and the AI disclosure within the srs document.
3. **The prompt(s) used:**
    - "All these files are related to the parking reservation app. Tell me what updates I should do on these files so that there are no discrepancies. Once done Follow the SRS template and place each of these files under the correct sections. For files that are pictures/google sheets write in bold red [INSERT <NAME OF TASK> HERE]. Do not change any of the wording from the files copy it exactly."
    - "can you make the document into a google doc instead?"
    - "[Pasted SRS template text for section 1.3] Whats the purpose of this"
    - "What else can be currently added to the srs document?"
    - "Do everything"
    - "[Pasted examples of REQ-FUNC-001, REQ-FUNC-002, REQ-FUNC-003] do the rest of the user stories in this format"
    - "Where should ai_use be on the srs document?"
    - "Where do i place all four tables in the srs?"
4. **What I kept vs. what I changed:**
    - **Verbatim / Kept:** I kept the requirement statements (REQ-FUNC-004 through REQ-FUNC-029 and the various REQ-SEC/REQ-PERF constraints) as the AI formatted them from my user stories. I also kept the new definitions provided for the Glossary and the AI's recommendations for tables (from homework 5) placements.
    - **Modified:** I manually modified Software-Corp SRS, making sure all the tables, Markdown formatting, and red placeholder text for images were aligned correctly. I also updated the Risk Register owner columns and user personas manually to resolve the discrepancies the AI initially pointed out.